

The Factorial-Denominator Series: Complete Reasoning Record

Erdős Problem #68

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ABSTRACT

Write $d_n = n! - 1$, $H_M = \sum_{2 \leq n \leq M} d_n^{-1}$ and $L_M = \text{lcm}(d_2, \dots, d_M)$. For a prime p with $e = v_p(L_M) \geq 1$, let $J = \{n : 2 \leq n \leq M, v_p(d_n) = e\}$ and write $d_n = p^e u_n$ on J . Specialising the reciprocal-sum valuation formula of Louwsma and Martino gives

$$v_p(\text{den}(H_M)) = e \iff \sum_{n \in J} u_n^{-1} \not\equiv 0 \pmod{p},$$

the inverses being taken modulo p . The prefixes through 138 and 2592 cancel the primes 139 and 2593 completely. The common denominator before reduction satisfies the quantitative bound

$$\liminf_{N \rightarrow \infty} \frac{\log L_N}{N^{3/2} \log N} \geq \frac{2\sqrt{2}}{3},$$

by an elementary segment argument that uses $\gcd(i! - 1, j! - 1) \mid j!/i! - 1$ and no external multiplicity theorem. The two statements concern opposite sides of reduction, and the cancellations at 139 and 2593 are why the second cannot be substituted for the first.

Wilson's theorem gives arbitrarily late first occurrences of primes among the factorial gaps, so prefix-private prime support of the denominators occurs cofinally, with no bound relating the prime to its first hit. Four exact criteria locate the irrationality of $S = \sum_{n \geq 2} d_n^{-1}$: a strict-successor carry equivalence, a companion-orbit residue condition, a lower-interval escape condition of width $O(m^{-2})$, and a boundary covering the whole shifted family $\sum_{n \geq 2} 1/(n! + t)$ for every integer $t \geq -1$, whose member $t = 0$ returns the irrationality of e . A finite-channel radius bound gives $3t^3 < 2(R + 1)$ under the exact cancellation and factorial-size hypotheses, and the growth statement above raises the asymptotic constant in that estimate from $3/2$ to $16/9$. Two finite computations exclude denominators: $q \nmid 299999!$ from an exact interval carry census through 300000, and $q \geq 2^{39990} > 10^{12038}$ from an exact continued-fraction enclosure computed outside Lean. The factorial-grid exclusion uses the external GMP computation and remains a separate outstanding Lean obligation. The irrationality of S is open, and the remaining arithmetic inputs are stated in the last section.

**Authorship and AI use.* Will Cook built and directed the research infrastructure and reviewed the claims when he could. AI agents did most of the research and drafting. Cook did not independently verify every claim.

Reading route and dependencies. For the main mechanism, read the survival criterion in §1, the lcm argument in §2, and the comparison in §4. For the strongest finite conclusions go directly to §6. The channel lattice in §5 is an independent explanation of a limitation, not an input to the finite computations. The research targets in §7 are explicitly conditional. The appendices retain longer deductions; source and execution status are recorded separately.

1 Which prime powers survive reduction

Problem 1.1 (Erdős #68). Is $S = \sum_{n \geq 2} (n! - 1)^{-1}$ irrational?

Erdős states the question on p. 102 of his 1988 survey and, in the same passage, records the expectation that $\sum_n 1/(n! + t)$ is irrational, indeed transcendental, for every integer t [1, p. 102]. That is conjectural context. Numbering follows [Bloom's Erdős problem catalogue](#) [2]. The catalogue reference retains the manuscript's historical access date; the current page could not be rechecked during this revision. No irrationality proof is supplied here.

Put

$$d_n = n! - 1, \quad H_M = \sum_{n=2}^M \frac{1}{d_n}, \quad L_M = \text{lcm}(d_2, \dots, d_M), \quad A_M = \sum_{n=2}^M \frac{L_M}{d_n},$$

so that $H_M = A_M/L_M$. Throughout, $\text{den}(x)$ is the positive denominator of a rational number in lowest terms. The first question is which prime power present in L_M remains in $\text{den}(H_M)$. For the largest exponent the answer is a specialisation of the reciprocal-sum valuation formula of Louwsma and Martino [4, Lemma 4.1, p. 10]; the proof below writes it out for the denominators d_n .

The general valuation mechanism is inherited. The explicit factorial specialisation, its cancellation witnesses, and its role in the scale comparison below are the contributions examined here; no priority claim for a new general reciprocal-sum valuation theorem is intended.

Theorem 1.2 (maximal prime-power survival). *Let $M \geq 2$, let p be a prime dividing L_M , and put $e = v_p(L_M)$. Let $J = \{n : 2 \leq n \leq M, v_p(d_n) = e\}$ and write $d_n = p^e u_n$ for $n \in J$. Then, with inverses in $\mathbb{F}_p$,*

$$v_p(\text{den}(H_M)) = e \iff \sum_{n \in J} u_n^{-1} \neq 0 \text{ in } \mathbb{F}_p. \quad (1)$$

Proof. Write $L_M = p^e W$ with $p \nmid W$. If $v_p(d_n) < e$, then $p \mid L_M/d_n$. If $n \in J$, then $u_n(L_M/d_n) = W$, so $L_M/d_n \equiv W u_n^{-1} \pmod{p}$. Summing over $2 \leq n \leq M$ gives

$$A_M \equiv \frac{L_M}{p^e} \sum_{n \in J} u_n^{-1} \pmod{p}. \quad (2)$$

Finally $\text{den}(H_M) = L_M / \gcd(A_M, L_M)$, whose p -valuation is e exactly when $p \nmid A_M$. Since W is invertible modulo p , that is exactly the stated condition. $\square$

The criterion decides the survival of the full exponent e . A zero residue gives $v_p(\text{den}(H_M)) < e$, and further congruences are needed to determine the surviving exponent. The proof uses no property of factorials beyond the displayed denominators, so it applies to any finite sum of reciprocals. What is specific here is the behaviour of the actual family $d_n = n! - 1$, and that behaviour is not generic.

Two complete cancellations. Take $M = p - 1$. The indices below with $p \mid d_n$ are all of them, and every one has valuation exactly one.

p	indices n	$d_n/p \pmod{p}$	inverses modulo p
139	69, 122, 137	6, 49, 73	116, 122, 40
2593	349, 2243, 2591	1508, 1566, 1678	1367, 356, 870

The inverse sums are $278 = 2 \cdot 139$ and 2593 . By Theorem 1.2, $139 \nmid \text{den}(H_{138})$ and $2593 \nmid \text{den}(H_{2592})$: each prime divides the common denominator of its prefix and vanishes from the reduced one. Both hit lists and both valuations are checked by the recurrence

$$r_1 = 1, \quad r_n \equiv nr_{n-1} \pmod{p^2}, \quad 0 \leq r_n < p^2,$$

run through $n = p - 1$. A hit is an index with $r_n \equiv 1 \pmod{p}$, its lifted cofactor is $(r_n - 1)/p$ modulo p , and valuation two would mean $r_n = 1$, which occurs at neither prime among the tested indices $2 \leq n \leq p - 1$. That enumeration is a finite calculation separate from the theorem.

The recurrence is run modulo p^2 , not merely modulo p : reduction modulo p would locate hits but could not establish that their valuations are exactly one. Both examples concern complete cancellation because the maximal exponent is one.

A prime is *prefix-private at m* when it divides d_m and divides none of $d_2, \dots, d_{m-1}$.

The word private is relative to that prefix, not to all future factorial gaps. Later hits can destroy uniqueness in a larger block.

Proposition 1.3 (cofinal first prime occurrences). *For every integer $B \geq 0$ there are a prime q and an integer $m > B$ with $m < q$, $q \mid m! - 1$ and $\gcd(q, k! - 1) = 1$ for every k with $2 \leq k < m$.*

Proof. Choose a prime $q \geq B! + 5$. Wilson's theorem gives $(q - 2)! \equiv 1 \pmod{q}$, so the set of indices $n \geq 2$ with $q \mid n! - 1$ is nonempty and contains no element above $q - 2$; let m be its least element. If $m \leq B$, then $q \leq m! - 1 \leq B! - 1$, contradicting the choice of q . Hence $m > B$, and minimality gives the coprimality assertions. $\square$

The construction gives no useful upper bound for q in terms of its first hit, and that comparison is what the scale inequalities of §4 consume. Wilson reflection limits what can be inferred from a large prime factor alone: for odd $n < q$ with $q \mid n! - 1$, the identity $(q - 1 - n)!n! \equiv (-1)^{n+1} \pmod{q}$ gives the reflected hit $q - n - 1$, which differs from n when $q \neq 2n + 1$ and is at least 2 when $n \leq q - 3$. The reflection identity is classical; see Stewart [8, p. 462, (4)].

Status. The problem treated here is open, and this note does not close it. Every statement below marked as checked is a proposition that the pinned Lean kernel accepts

from the sources this note links to, with no sorry, no added axiom, and no unchecked evaluation. That is a claim about the formal statement, not about its mathematical interest, its novelty, or the original problem. The unresolved obligations are named exactly, in their own section, and none of the finite computations, reductions, or no-go results here removes one of them.

Companion system context. The [claim and trust boundary](#), [cold-clone route to proof authority](#), and [public contribution protocol](#) are described in sibling papers. Those descriptions do not change the mathematical status of this note.

2 The growth of the common denominator

Theorem 1.2 concerns the denominator after reduction. The common denominator before reduction admits an unconditional lower bound of its own, by an elementary argument.

Lemma 2.1 (product, least common multiple, pairwise gcd). *For positive integers $x_1, \dots, x_k$,*

$$\prod_{i=1}^k x_i \mid \text{lcm}(x_1, \dots, x_k) \prod_{i < j} \gcd(x_i, x_j).$$

Proof. Fix a prime r and relabel so that $a_1 \leq \dots \leq a_k$, where $a_i = v_r(x_i)$. The right-hand side has r -valuation $a_k + \sum_{i < j} \min(a_i, a_j) = a_k + \sum_{i=1}^{k-1} (k-i)a_i$, and the left-hand side has $\sum_{i=1}^k a_i$. The difference is $\sum_{i=1}^{k-1} (k-i-1)a_i$, which is nonnegative. $\square$

The next divisibility is the case $P = -1$ of the relation $\gcd(i! + P(i), j! + P(j)) \mid (j!/i!)P(i) - P(j)$ for $P \in \mathbb{Z}[X]$, obtained by multiplying $i! + P(i)$ by $j!/i!$ and subtracting $j! + P(j)$. Luca and Shparlinski use this relation to bound a common divisor of $n! + P(n)$ and $(n+h)! + P(n+h)$ in the proof of their Lemma 5 [13], and Lai uses it for a common prime-power divisor [14, proof of Lemma 2.4, display (2.5)].

The earlier spacing method of Erdős and Stewart [18, §3, pp. 516–517] is related background. We now state only the subtraction needed for the lcm argument, rather than importing the prime-factor estimates of those papers.

Lemma 2.2 (factorial-gap gcd). *For $2 \leq i < j$, the integer $g = \gcd(i! - 1, j! - 1)$ divides $j!/i! - 1$, and $g \leq j!/i! - 1 < j^{j-i}$.*

Proof. Both $i!$ and $j!$ are congruent to 1 modulo g . The quotient $j!/i! = (i+1)(i+2)\dots j$ is an integer, and $j! = i! \cdot (j!/i!)$, so $j!/i! \equiv 1 \pmod{g}$. Since $j!/i! \geq i+1 > 1$, the positive integer $j!/i! - 1$ is a multiple of g , whence $g \leq j!/i! - 1$. Finally $j!/i!$ is a product of $j-i$ integers each at most j . $\square$

Lemma 2.3 (segment inequality). *For $2 \leq k \leq N-1$,*

$$\sum_{n=N-k+1}^N \log(n! - 1) \leq \log L_N + \binom{k+1}{3} \log N. \quad (3)$$

Proof. Apply Lemma 2.1 to $x_n = d_n$ for $N - k + 1 \leq n \leq N$. Their least common multiple divides L_N . By Lemma 2.2 each pairwise gcd is smaller than N^{j-i} , and over a block of k consecutive indices

$$\sum_{i < j} (j - i) = \sum_{d=1}^{k-1} d(k-d) = \binom{k+1}{3}.$$

Taking logarithms of the resulting divisibility gives (3). □

Theorem 2.4 (common-denominator growth).

$$\liminf_{N \rightarrow \infty} \frac{\log L_N}{N^{3/2} \log N} \geq \frac{2\sqrt{2}}{3}.$$

Proof. A terminal block keeps every factorial near $N!$, but increasing its length also increases the total pairwise-gcd loss. The gain is of order $kN \log N$ and the loss of order $k^3 \log N$, so the useful balance is k of order $\sqrt{N}$. Fix $\alpha > 0$ and take $k = \lfloor \alpha \sqrt{N} \rfloor$, which satisfies $2 \leq k \leq N - 1$ for all large N . Put $u = N - k + 1$.

For the left-hand side of (3), use $n! - 1 \geq n!/2$ and $\log n! \geq n \log n - n$. Since $n \mapsto n \log n - n$ increases,

$$\sum_{n=u}^N \log(n! - 1) \geq k(u \log u - u) - k \log 2.$$

Here $u \geq N - \alpha \sqrt{N}$ and $k = \alpha \sqrt{N} + O(1)$, so $k u \log u = \alpha N^{3/2} \log N + O(N \log N)$, while $k u = O(N^{3/2})$ and $k \log 2 = O(\sqrt{N})$. Hence the left-hand side is at least $\alpha N^{3/2} \log N + o(N^{3/2} \log N)$.

For the right-hand side, $\binom{k+1}{3} \leq (k+1)^3/6$, so $\binom{k+1}{3} \log N = \frac{\alpha^3}{6} N^{3/2} \log N + o(N^{3/2} \log N)$. Therefore

$$\liminf_{N \rightarrow \infty} \frac{\log L_N}{N^{3/2} \log N} \geq \alpha - \frac{\alpha^3}{6}.$$

The right-hand side is maximised at $\alpha = \sqrt{2}$, with value $\sqrt{2} - \frac{2\sqrt{2}}{6} = \frac{2\sqrt{2}}{3}$. □

The displayed proof of the lcm bound uses only Lemmas 2.1 and 2.2. Its formal counterpart is recorded in the source ledger; the status of that entry is not a blanket assertion about all linked modules. Theorem 12 of Garaev, Luca and Shparlinski [7, arXiv v1, p. 16] supplies a different input: a uniform $O(N^{2/3})$ bound for a fixed nonzero factorial fibre in an interval of length N below a prime modulus. The weaker lcm deduction from it is retained in §B.3, not used here. The revision makes no exhaustive priority claim for the lcm estimate; its precise elementary derivation and the inherited subtraction are stated so that the claim can be assessed without relying on a search conclusion.

Remark (polynomial shifts). Fix $P \in \mathbb{Z}[X] \setminus \{0\}$. By Lemma 3 of Luca and Shparlinski [13], in the form given by Lai [14, Lemma 2.1], there is $n_0 \geq 2$, depending only on P ,

such that $n! + P(n) > 1$ for all $n \geq n_0$ and $P(n)(n+1) \cdots (n+h) \neq P(n+h)$ for all $n \geq n_0$ and $h \geq 1$. The proof of Theorem 2.4 then gives

$$\liminf_{N \rightarrow \infty} \frac{\log \text{lcm}\{n! + P(n) : n_0 \leq n \leq N\}}{N^{3/2} \log N} \geq \frac{2\sqrt{2}}{3}.$$

For $n_0 \leq i < j \leq N$, multiplying $i! + P(i)$ by $j!/i!$ and subtracting $j! + P(j)$ gives

$$\gcd(i! + P(i), j! + P(j)) \mid \frac{j!}{i!} P(i) - P(j),$$

which replaces Lemma 2.2. The right-hand side is nonzero by the choice of n_0 , and its absolute value is at most $C_P N^{j-i+\deg P}$ for a constant C_P . Over a block of k indices the pairwise loss therefore grows by $O_P(k^2 \log N)$, which is $O_P(N \log N)$ when $k = \lfloor \alpha \sqrt{N} \rfloor$. Since $n! + P(n) \geq n!/2$ for large n , the lower estimate for the sum of logarithms is unchanged. The cutoff is needed, since for $P = -2$ the factor at $n = 2$ vanishes. The hypothesis $P \neq 0$ is needed as well, since for $P = 0$ the least common multiple is $N!$ whatever the cutoff.

A non-polynomial comparison. For $n! + 2^n - 1$, Luca and Shparlinski [19, Lemmas 2.1–2.3, pp. 860–862] eliminate the exponential term using three hits; multiplicative order enters the counting. Their valuation-layer identity [19, (3.2), p. 863] records repeated prime-power divisibility. These are related tools, not inputs to Theorem 2.4, and no extension to general non-polynomial perturbations is asserted.

Within the terminal-block estimate just proved, maximising $\alpha - \alpha^3/6$ gives $2\sqrt{2}/3$. This is an optimisation of that one-parameter bound, not an optimality theorem for all block selections or all gcd arguments. The discarded factorial term has size $N^{3/2}$, so a finite normalised ratio need not be close to the limiting lower bound.

Theorem 2.4 is a statement about L_N and not about $\text{den}(H_N)$. Consecutive denominators $(N-1)! - 1$ and $N! - 1$ are already coprime for $N \geq 3$, so the qualitative failure of common-denominator clearing follows from two terms; the estimate is the quantitative growth statement. Theorem 1.2 and the cancellations at 139 and 2593 show why the substitution fails: a prime may occupy the common denominator and be absent from the reduced one. One cancelled prime is not an asymptotic cancellation theorem.

For completeness, the consecutive coprimality follows because their gcd divides $N-1$, whereas $(N-1)! - 1$ is coprime to $N-1$. Thus the two-term argument is qualitative and does not depend on the asymptotic constant.

The direction of denominator control matters. Write $H_N = \hat{A}_N/Q_N$ in lowest terms, reserving A_N above for the numerator over the full common denominator. Under the hypothetical identity $S = a/q$, with $a \in \mathbb{Z}$ and $q \geq 1$, positivity of the tail gives

$$qQ_N(S - H_N) = aQ_N - q\hat{A}_N \in \mathbb{Z}_{>0}.$$

A direct clearing contradiction therefore requires an upper estimate that makes $qQ_N(S - H_N) < 1$ for some N ; equivalently, it needs Q_N to be small relative to the reciprocal tail. A lower bound on Q_N points in the opposite direction. Indeed, the rational limit 1 and the approximants $1 - 1/Q_N$ have arbitrarily large reduced denominators without any

irrationality. The inequality $d_N(S - H_N) < 1$ alone would not repair the argument, because d_N generally does not clear the earlier summands of H_N . The prime-pole analysis describes which factors survive reduction, but it does not yet supply the required upper control on Q_N .

For linear forms in p -adic zeta values, with $p \geq 5$ a fixed prime and n the index of the form, Lai, Lupu and Sprang quantify a saving of this kind. They bound the denominators of the coefficients by powers of $\text{lcm}(1, \dots, n)$, show that the bound may be divided by a product Φ_n of prime powers, and compute the growth rate of Φ_n [15, Lemmas 5.3–5.7 and 7.3]. That saving enters the inequality [15, (8.1)] under which the rescaled forms, which have integer coefficients, meet the irrationality criterion [15, Lemma 2.1, p. 3] that they quote from Lai [16, Lemma 2.1, p. 4]: nonzero values whose p -adic size, multiplied by the largest absolute value of a coefficient, tends to zero. The corresponding saving for H_N is the ratio L_N/Q_N .

This ratio is an integer. Merely naming it supplies neither its growth nor a small linear form; the analogy is about what an eventual estimate would have to accomplish. Theorem 1.2 decides, for each prime $p \mid L_N$, whether p divides this ratio; no asymptotic lower bound for the ratio is established here. This paragraph compares the two methods; no p -adic theorem is applied to S .

3 Carries and the rationality boundary

Define the strict integer successor and the predecessor gap by

$$Z_m = \lfloor m!H_m \rfloor + 1, \quad \Delta_m = Z_{m-1} - (m-1)!H_{m-1} \quad (m \geq 3),$$

so $0 < \Delta_m \leq 1$, and define the integer carry b_m by

$$Z_m = mZ_{m-1} + 1 - b_m. \quad (4)$$

Call $b_m = 1$ a *unit carry*. Put $E_m = m!(S - H_m)$ and $\varepsilon_m = 1/(m! - 1)$. Since $d_{n+1} > (n+1)d_n$, a geometric majorant gives the tail estimate

$$0 < E_m < \frac{2m!}{(m+1)! - 1} < \frac{2}{m} \leq 1 \quad (m \geq 2). \quad (5)$$

The proof of the next theorem feeds this estimate into the factorial-tail integrality argument that Hančl and Tijdeman use for factorial series with integer coefficients [11, Lemma 2.1 and the following remark, p. 385].

Theorem 3.1 (strict-successor characterisation). *For $m \geq 3$,*

$$b_m = 1 \iff m \mid Z_m \iff 1 + \varepsilon_m < m\Delta_m \leq 2 + \varepsilon_m. \quad (6)$$

The historical manuscript cites Erdos68/PaperCompleteExisting.lean, line 79. This module is absent from the retained source pin; this citation does not establish kernel verification at that pin.

Moreover

$$S \in \mathbb{Q} \iff b_m = 1 \text{ for all sufficiently large } m, \quad (7)$$

$$S \notin \mathbb{Q} \iff \forall B \exists m > B : m \nmid Z_m. \quad (8)$$

If $S = a/q$ with $a \in \mathbb{Z}$, $q \geq 1$ and $b_m \neq 1$, then $q \nmid (m-1)!$ and $q \geq m$.

Proof. From $m!H_m = mZ_{m-1} - m\Delta_m + 1 + \varepsilon_m$ one gets $b_m = \lceil m\Delta_m - 1 - \varepsilon_m \rceil$ with $-1 \leq b_m \leq m-1$, which gives both equivalences in (6), endpoints included.

Suppose $S = a/q$ and $q \mid (m-1)!$. For $j = m-1$ and $j = m$ the number $j!S$ is an integer, and $j!H_j = j!S - E_j$ with $0 < E_j < 1$ by (5), so $Z_j = j!S$. Substituting into (4) forces $b_m = 1$. Every fixed q divides $(m-1)!$ eventually, which gives one direction of (7) and, by contraposition, the final assertion; $q < m$ would imply $q \mid (m-1)!$. Conversely, an eventual unit-carry tail makes $Z_m/m!$ eventually constant, and $0 < Z_m/m! - H_m \leq 1/m!$ with $H_m \rightarrow S$ identifies that constant as S , which is then rational. Negating (7) gives (8). $\square$

The divisibility conclusion retains prime-power multiplicities, so it constrains valuations, prime support included. It does not require a prime divisor of q larger than $m-1$.

3.1 Why one proposed window argument is circular

One route to cofinal non-unit carries tries to exclude two consecutive unit carries by extracting a prime-power or quotient-gcd obstruction from their cleared width-one window. The exact two-step recurrence shows what that window can deliver. Let U_k and V_k be the numerator and denominator of the reduced predecessor gap and let $G_k > 0$ be the exact transition normaliser. For $m \geq 3$, the two unit carries are equivalent to an integer offset Ω_m with $0 < \Omega_m \leq D_m$, the window denominator telescopes independently of the carry values,

$$D_m = V_{m+2}G_{m+1}G_m = V_m(m! - 1)((m+1)! - 1),$$

and under the pair assumption the offset factors to match, $\Omega_m = U_{m+2}G_{m+1}G_m$. Cancelling the common positive normaliser reduces the complete window to

$$0 < U_{m+2} \leq V_{m+2},$$

which every reduced positive gap in $(0, 1]$ already satisfies. For that specified window, therefore, an obstruction obtained only after imposing the carry pair rewrites a bound that holds anyway. The calculation does not disprove adjacent unit carries and does not exclude an independent restriction on the transition numerators or their valuations, which would need its own argument constraining the predecessor state before the pair is assumed.

3.2 The companion coordinate

Let $C = \sum_{n \geq 2} (n!(n! - 1))^{-1}$. The identity $1/(n! - 1) = 1/n! + 1/(n!(n! - 1))$ and absolute convergence give

$$S = C + e - 2. \tag{9}$$

The canonical factorial digits of a real x are $a_m(x) = \lfloor m!x \rfloor - m \lfloor (m-1)!x \rfloor \in \{0, \dots, m-1\}$, and x is rational exactly when they vanish from some index on. The criterion goes back to Cantor [5]; Galambos treats the rationality of Cantor series in [6, Ch. II, §2.1, pp. 21–22], and Koepf and Schmiersau prove the irrationality direction for nonterminating factorial expansions whose digits are not eventually maximal [9, Example 3.2, p. 121].

Proposition 3.2 (companion-orbit criterion).

$$S \in \mathbb{Q} \iff \lfloor m!C \rfloor \equiv -2 \pmod{m} \text{ for all sufficiently large } m.$$

Proof. Put $J_m = \sum_{k=0}^m m!/k!$. Then J_m is an integer, every summand with $k < m$ is divisible by m and the summand at $k = m$ is 1, so $J_m \equiv 1 \pmod{m}$; and $0 < m!e - J_m < 1$ for $m \geq 2$.

Suppose S is rational. For all large m both $(m-1)!S$ and $m!S$ are integers, so $m \mid m!S$. Multiplying (9) by $m!$ and using $\lfloor -m!e \rfloor = -J_m - 1$ gives $\lfloor m!C \rfloor = m!S + 2m! - J_m - 1 \equiv -2 \pmod{m}$.

Conversely, assume the congruence from some index on. Because $0 \leq a_m(C) < m$, for $m \geq 3$ the congruence is equivalent to $a_m(C) = m - 2$. Choose N beyond the exceptional indices. The canonical expansion gives

$$C = \frac{\lfloor N!C \rfloor}{N!} + \sum_{m>N} \frac{m-2}{m!},$$

and the telescope $\sum_{m>N} (m-1)/m! = 1/N!$ turns this into

$$S = C + e - 2 = \frac{\lfloor N!C \rfloor + 1}{N!} + \sum_{m=2}^N \frac{1}{m!},$$

which is rational. □

3.3 Escape from a smaller interval

Write $\theta_m = \{m!S\}$. The exact lower interval has width $E_m/m \in (0, 2/m^2)$, so the target shrinks quadratically.

Proposition 3.3 (lower-interval criterion).

$$S \notin \mathbb{Q} \iff \forall B \exists m > B : E_m \leq m\theta_{m-1}. \quad (10)$$

For $m \geq 3$ the finite condition

$$m\Delta_m \leq 1 + \varepsilon_m \quad \text{or} \quad 1 + \varepsilon_m + \frac{2}{m} \leq m\Delta_m \quad (11)$$

implies the escape inequality in (10). Cofinally many instances of (11) therefore imply $S \notin \mathbb{Q}$.

Proof. If S is rational, the remainders θ_{m-1} vanish eventually while $E_m > 0$, so the escape inequality fails eventually. Conversely, suppose $m\theta_{m-1} < E_m$ for every sufficiently large m . By (5) this gives $m\theta_{m-1} < 1$, so $a_m(S) = 0$ and $\theta_m = m\theta_{m-1}$. A positive remainder grows past one under that recurrence, so the remainders vanish and S is rational.

For the finite implication, suppose $m\theta_{m-1} < E_m$. The tail recurrence is $mE_{m-1} = 1 + \varepsilon_m + E_m$, so $E_{m-1} > E_m/m > \theta_{m-1} \geq 0$, and with $0 < E_{m-1} < 1$ the strict-successor identity gives $\Delta_m = E_{m-1} - \theta_{m-1}$. Hence

$$1 + \varepsilon_m < m\Delta_m = 1 + \varepsilon_m + E_m - m\theta_{m-1} \leq 1 + \varepsilon_m + E_m < 1 + \varepsilon_m + \frac{2}{m},$$

which contradicts (11). □

Escape at a single index does not give a non-unit carry there. The exact classification is

$$E_m \leq m\theta_{m-1} \iff b_m \neq 1 \text{ or } a_m(S) = m - 1 \quad (m \geq 3), \quad (12)$$

The historical manuscript cites Erdős68/PaperCompleteExisting.lean, line 104. This module is absent from the retained source pin; this citation does not establish kernel verification at that pin.

so the maximal-digit branch is available. It is realised: at $m = 52$ exact rational arithmetic gives $b_{52} = 1$ together with $52\Delta_{52} > 1 + \varepsilon_{52} + 2/52$, the margin being about 0.5689674908. Since 52 is composite, this settles nothing about prime indices, and the cofinal statement in Proposition 3.3 does not identify escape with a non-unit carry.

3.4 The whole shifted family

Erdős asked about S inside a family, and the same coordinate covers the family. For an integer $t \geq -1$ put $S_t = \sum_{n \geq 2} 1/(n! + t)$ and $C_t = \sum_{n \geq 2} 1/(n!(n! + t))$, so that $S_t = -tC_t + (e - 2)$.

The restriction $t \geq -1$ keeps every denominator positive for $n \geq 2$. All series in this identity converge absolutely.

Theorem 3.4 (uniform family boundary). *For every integer $t \geq -1$, the series S_t is rational exactly when*

$$[tm!C_t] \equiv 2 \pmod{m}$$

for all sufficiently large m , and irrational exactly when that residue is missed cofinally. The member $t = -1$ is S , and the member $t = 0$ is $e - 2$, whose orbit is constantly 0 and therefore misses the residue class at every $m \geq 3$, so the boundary returns the irrationality of e .

Proof. Put $Y = -tC_t$, so that $S_t = Y + e - 2$. The proof of Proposition 3.2 applies to any real Y : rationality of $Y + e - 2$ forces $[m!Y] \equiv -2 \pmod{m}$ eventually. Conversely, that congruence forces the canonical digits of Y to equal $m - 2$ eventually; adding the factorial series of $e - 2$ leaves an eventually telescoping tail, hence a rational sum. Finally $[tm!C_t] = -[m!Y]$ translates the congruence. Negation gives the cofinal statement. For $t = 0$, the left side is zero, which is not congruent to 2 for $m \geq 3$. $\square$

The $t = 0$ specialisation recovers the classical irrationality of e . For $t \neq 0$ this paper supplies no cofinal estimate for the displayed orbit. This is a statement about the scope of the present argument, not an assertion that every other member has the same current literature status.

4 From denominator structure to a scale criterion

For an integer $p \geq 3$ put $I_p = \{2, \dots, 2p - 1\}$ and $F_p = (p - 1)!$, and define

$$\begin{aligned} D_p &= \text{lcm}_{\substack{i,j \in I_p \\ i < j}} \gcd(d_i, d_j), & C_p &= \text{lcm}(F_p, D_p), \\ L_p^{\text{blk}} &= \text{lcm}(F_p, d_2, \dots, d_{2p-1}), & R_p &= L_p^{\text{blk}}/C_p. \end{aligned}$$

The normalised core is $\tilde{C}_p = C_p/F_p$, so that

$$L_p^{\text{blk}} = C_p R_p = F_p \tilde{C}_p R_p. \quad (13)$$

The historical manuscript cites Erdős68/PaperCompleteExisting.lean, line 125. This module is absent from the retained source pin; this citation does not establish kernel verification at that pin.

The factorial base F_p belongs inside every one of these quantities, and omitting it changes the scale comparisons below by a factorial factor. Put

$$T_p = \sum_{i \in I_p} \frac{L_p^{\text{blk}}}{d_i}, \quad \rho_p = (-T_p) \bmod R_p, \quad K_p = 2p^2(2p-1)!,$$

with ρ_p the least nonnegative representative. All of these depend only on a finite prefix.

In this section p is a natural parameter, not necessarily a prime. Prime divisors used in the valuation argument are separate variables. The distinction matters when the parameter is tailored to a first hit.

The core records every prime-power layer occurring in at least two gaps, together with the base. A prime dividing R_p therefore has a unique maximal-valuation gap above the base valuation, so the argument of Theorem 1.2 applies to it and gives $\gcd(T_p, R_p) = 1$. This is where the first theorem returns. Consequently, when $R_p > 1$, the ratio $T_p/R_p = C_p H_{2p-1}$ is not an integer and

$$\frac{\rho_p}{R_p} = \lceil C_p H_{2p-1} \rceil - C_p H_{2p-1} \in (0, 1), \quad (14)$$

a gap to the next integer.

Theorem 4.1 (complementary-residue criterion). *Suppose that for every B there is a natural parameter $p \geq 3$ with $p > B$, $R_p > 1$, and*

$$(2p+1)L_p^{\text{blk}} < K_p \rho_p. \quad (15)$$

Then S is irrational.

This natural-parameter form is also the [source form of the global complementary criterion](#).

Proof. Suppose $S = a/q$ with $q \geq 1$, and take such a parameter with $p > \max(q, 3)$ satisfying the hypotheses. Then $q \mid F_p$, so $U = L_p^{\text{blk}}(S - H_{2p-1})$ is a positive integer, and $L_p^{\text{blk}}S = R_p(C_p S)$ is a multiple of R_p ; hence $U \equiv -T_p \pmod{R_p}$ and $U \geq \rho_p$. For the other side, $1/(n-1) < 2/n!$ for $n \geq 3$, and a geometric majorant gives

$$0 < S - H_{2p-1} < \frac{2}{(2p)!} \cdot \frac{2p+1}{2p} = \frac{2p+1}{K_p}.$$

Therefore $K_p \rho_p \leq K_p U < (2p+1)L_p^{\text{blk}}$, contrary to (15). $\square$

Cancelling R_p in (15) through (13) displays the quantitative issue directly:

$$(2p+1)C_p < K_p \frac{\rho_p}{R_p}. \quad (16)$$

The private modulus has gone. A supply of large private factors therefore supplies neither a small collision core nor a lower bound for the normalised gap ρ_p/R_p , which is at most one. That is the exact reason Proposition 1.3 does not close the argument.

Writing $\eta_p = \rho_p/R_p$, the sufficient comparison becomes

$$\log \tilde{C}_p - \log \eta_p < \log \frac{2p^2(2p-1)!}{(2p+1)(p-1)!} = p \log p + (2 \log 2 - 1)p + O(\log p).$$

It is a joint budget for collision loss and the real gap. The bound $\rho_p \geq 1$ gives only $\eta_p \geq 1/R_p$ and may be far too small.

The moving-factor criterion is a more specialised sufficient condition. For a prefix-private prime q at $m \geq 4$ the tailored block parameter is $p = \lfloor m/2 \rfloor + 1$, which need not itself be prime. For the factors 1 and q of the private modulus the factor-pair floor is $\min(\rho_p, R_p/q)$, and its scale inequality splits exactly:

$$(2p+1)L_p^{\text{blk}} < K_p \min(\rho_p, R_p/q) \iff \begin{cases} (2p+1)L_p^{\text{blk}} < K_p \rho_p, \\ (2p+1)C_p q < K_p. \end{cases} \quad (17)$$

Testing both entries of the minimum and using (13) gives the equivalence, so the reduction has no opaque floor premise left. Cofinal instances of both inequalities imply irrationality. The single global condition of Theorem 4.1 is separately sufficient, so the pair is one sufficient route among several.

One structural limitation is worth recording beside these criteria. If the owner index n is fixed and $p > n! - 1$, then $n! - 1$ divides $(p-1)!$, so its private quotient in the block at p is exactly one. Large private quotients seen at small blocks, such as the factor 719 owned at $n = 6$, are therefore finite-range phenomena, and the selected factors must escape with p .

5 A limitation of finite-channel cancellation

For a finitely supported integer vector $c = (c_i)$ on indices $i \geq 2$, put

$$M(c) = \sum_i c_i i!, \quad W_{d,i} = \frac{i!}{(d!)^{\lfloor i/d \rfloor}}, \quad V_d(c) = \sum_i c_i W_{d,i} \quad (d \geq 2).$$

Each $W_{d,i}$ is an integer: writing $i = kd + r$ with $0 \leq r < d$, the quotient $i!/((d!)^k r!)$ is a multinomial coefficient. Since $i! = (d!)^{\lfloor i/d \rfloor} W_{d,i}$ and $d! \equiv 1 \pmod{d! - 1}$,

$$V_d(c) \equiv M(c) \pmod{d! - 1}. \quad (18)$$

Theorem 5.1 (integral normal form). *For every finite integer support and every $d \geq 2$ there is an integer k with $V_d(c) = M(c) + (d! - 1)k$.*

Proof. By (18), $d! - 1$ divides $V_d(c) - M(c)$. Thus $k = (V_d(c) - M(c)) / (d! - 1)$ is an integer and gives the identity. The modulus is positive also at $d = 2$, when it equals one. $\square$

Theorem 5.2 (quotient-band breakpoint). *Let $d \geq 2$ and $k \geq 0$, and suppose every supported index i satisfies $kd \leq i < (k+1)d$. Then $M(c) = (d!)^k V_d(c)$. In particular, cancellation in the first band $d \leq i < 2d$ forces $M(c) = 0$; and if every supported index is at least d while $M(c) \neq 0$ and $V_d(c) = 0$, then some supported index is at least $2d$.*

Proof. Within one band the quotient $[i/d]$ is the constant k , so $i! = (d!)^k W_{d,i}$ for every supported index and the factor $(d!)^k$ comes out of the sum. The two consequences follow by taking $k = 1$ and by contraposition. $\square$

This is a direct consequence of the constant quotient; no valuation estimate enters. A nonzero moment therefore cannot be hidden entirely below $2d$ while the d -channel cancels. This is a finite-family obstruction: it constructs no cancelling family and says nothing about simultaneous channels or residual size.

Consequently a vanishing d -th channel forces $(d! - 1) \mid M(c)$, and annihilating every channel $2 \leq d \leq D$ forces $L_D \mid M(c)$, where $L_D = \text{lcm}_{2 \leq d \leq D} (d! - 1)$ is the same quantity as in §2. Every zero-moment variation of the support changes a normalised channel contribution by an integer only, so such variations cannot manufacture an extra fractional cancellation coordinate.

5.1 The exact low-channel lattice and residual class

Fix $D \geq 2$. The low-channel equations, the support equation and the residual identity answer different questions: solvability, admissibility on $n \geq 2$, and the remaining fractional obstruction, respectively.

The integral divisor basis makes the preceding divisibility statement exact. For the coordinate calculation only, enlarge the coefficient space to finitely supported integer vectors on $n \geq 1$, with M and $V_d(\cdot)$ given by the same formulas; this introduces no term $1/(1! - 1)$ into S . Set

$$T_n = ne_{n-1} - e_n, \quad U_n = T_n - \sum_{\substack{d \mid n \\ 2 \leq d < n}} W_{d,n} U_d \quad (n \geq 2), \quad K_D = L_D e_1 - \sum_{d=2}^D \frac{L_D}{d! - 1} U_d.$$

The quotient $[i/d]$ changes between $i = n - 1$ and $i = n$ exactly when $d \mid n$, so $V_d(T_n) = (d! - 1)W_{d,n}\mathbf{1}_{d \mid n}$, and induction gives $M(U_n) = 0$ and $V_d(U_n) = (d! - 1)\mathbf{1}_{d=n}$. Hence $M(K_D) = L_D$ and $V_d(K_D) = 0$ for $2 \leq d \leq D$. The change from $e_1, T_2, T_3, \dots$ to $e_1, U_2, U_3, \dots$ is integral triangular with unit diagonal, so $e_1, U_2, U_3, \dots$ is an integral basis, and applying M and each $V_d(\cdot)$ gives the unique expansion

$$c = M(c) e_1 + \sum_{d \geq 2} \frac{V_d(c) - M(c)}{d! - 1} U_d.$$

Consequently every vector of the enlarged space whose channels $2, \dots, D$ vanish has a unique expression

$$c = tK_D + \sum_{n > D} z_n U_n, \quad M(c) = tL_D, \quad (19)$$

with finitely many nonzero integers z_n . Write $a_D = [e_1]K_D$ and $u_n = [e_1]U_n$. Requiring support on $n \geq 2$ adds precisely the scalar equation

$$ta_D + \sum_{n>D} z_n u_n = 0. \quad (20)$$

The scalar coefficients satisfy $u_2 = 2$ and $u_n = -\sum_{d|n, 2 \leq d < n} W_{d,n} u_d$ for $n > 2$; in particular $u_{2p} = -(2p)!/2^{p-1}$ for every prime p . If p is a prime with $D/2 < p \leq D$ and $H = D(2p-1)$, then

$$\gcd\{u_n : n > D\} = \gcd(u_{D+1}, \dots, u_H) > 0, \quad H < 2D^2. \quad (21)$$

Indeed, the finite gcd g divides $u_{2p} \neq 0$ and hence $(2p)!$. For $n > H$ and a divisor $d \leq D$ of n one has $n/d \geq 2p$, and $(n/d)!$ divides $W_{d,n}$, the quotient counting partitions of n points into n/d unordered blocks of size d ; thus $g \mid W_{d,n}$. Strong induction in the recurrence handles the remaining divisors $D < d < n$, so g divides every u_n with $n > D$. Bertrand's postulate supplies p for $D \geq 3$, and $p = 2$ serves for $D = 2$; finally $H \leq D(2D-1) < 2D^2$. The short note proves the same determination as its Theorem 3.2.

Theorem 5.3 (exact attainable moment ideal). *Fix $D \geq 2$ and a prime p with $D/2 < p \leq D$. Put*

$$H = D(2p-1), \quad G_D = \gcd\{|u_n| : D < n \leq H\}, \quad \mu_D = L_D \frac{G_D}{\gcd(G_D, a_D)}.$$

The moments of finite integer vectors supported on $n \geq 2$ and cancelling all channels $2, \dots, D$ are exactly $\mu_D \mathbb{Z}$. The integer μ_D is positive, is independent of the eligible prime p , and is attained by a vector of coefficient content one, hence by a primitive vector.

Proof. By (21), $G_D > 0$ is the gcd of the whole tail $\{u_n : n > D\}$, and $H < 2D^2$. Thus the finite integer combinations in (20) form $G_D \mathbb{Z}$, so that the equation is soluble exactly when $G_D \mid ta_D$. Dividing by $\gcd(G_D, a_D)$ shows that t is a multiple of $G_D / \gcd(G_D, a_D)$, and (19) gives the displayed moment ideal. Bezout coefficients attain its positive generator. If an attaining vector had a nontrivial common coefficient divisor, division by that divisor would produce a smaller positive attainable moment. Hence its content is one. The ideal itself is intrinsic, so two eligible primes give the same positive generator. $\square$

For

$$\mathcal{R}(c) = \sum_{d \geq 2} \frac{V_d(c)}{d! - 1}, \quad H_D = \sum_{2 \leq d \leq D} \frac{1}{d! - 1},$$

the same coordinates retain the real information that the moment ideal alone does not see.

Theorem 5.4 (full residual transparency). *For the vector in (19),*

$$\mathcal{R}\left(tK_D + \sum_{n>D} z_n U_n\right) = tL_D(S - H_D) + \sum_{n>D} z_n.$$

The residual series converges for every finite vector supported away from index zero. A zero-moment vector has integral residual, and any two finite vectors with the same factorial moment have residuals differing by an integer.

Proof. For $d > D$, the K_D channel is L_D , while each U_n contributes one to its isolated residual coordinate. Termwise decomposition therefore gives the factorial-gap tail $L_D(S - H_D)$ plus the finite coordinate mass $\sum z_n$. The factorial-gap tail is summable and the coordinate correction has finite support, which proves convergence. In the full divisor-basis coordinates the coefficient of the e_1 column equals the factorial moment; when that moment is zero only the integral coordinate mass remains. Subtracting two equal-moment identities proves the last assertion. $\square$

The exact Lean endpoint packages are [primitive attainment of the moment-ideal generator](#) and [the equal-moment residual class](#). They determine the finite low-channel lattice and its residual class modulo $\mathbb{Z}$; they do not prove that the remaining real residual is nonzero along a cofinal family.

Theorem 5.5 (square-subsequence radius bound). *Let $t, M, R \in \mathbb{N}$ satisfy*

$$t \geq 2^{32}, \quad M > 0, \quad L_{2t^2} \mid M, \quad M < (R+1)! - 1.$$

Then $3t^3 < 2(R+1)$. Consequently no family satisfying these hypotheses for all sufficiently large t has $(R(t) + 1)/t^3 \leq 3/2$ eventually, and none has $R(t) = o(t^3)$. The historical manuscript cites Erdos68/PaperCompleteExisting.lean, line 236. This module is absent from the retained source pin; this citation does not establish kernel verification at that pin.

Proof. Put $D = 2t^2$, $k = 2t$ and $u = D - k + 1 = 2t^2 - 2t + 1$. Lemma 2.3 at $N = D$, with $n! - 1 \geq n!/2$ and $\log n! \geq n \log n - n$, and with $L_D \leq M < (R+1)!$, gives

$$2t(u \log u - u - \log 2) < (R+1) \log(R+1) + \binom{2t+1}{3} \log(2t^2). \quad (22)$$

Suppose $R+1 \leq \frac{3}{2}t^3$. Since $t \geq 16$ one has $\frac{15}{8}t^2 \leq u \leq 2t^2$ and $\log u \geq 2 \log t$, so the left side of (22) is at least $\frac{15}{2}t^3 \log t - 4t^3 - 2t \log 2$. Using $\binom{2t+1}{3} < \frac{4}{3}t^3$ and $\log \frac{3}{2} < \log 2$, the right side is at most

$$\frac{3}{2}t^3(\log 2 + 3 \log t) + \frac{4}{3}t^3(\log 2 + 2 \log t).$$

Dividing by t^3 and rearranging, (22) would require

$$\frac{1}{3} \log t < 4 + \frac{2 \log 2}{t^2} + \frac{17}{6} \log 2.$$

But $\log t \geq 32 \log 2$ and $\frac{2}{3} < \log 2 < 1$ give $\frac{1}{3} \log t - \frac{17}{6} \log 2 \geq \frac{47}{6} \log 2 > 5$, while $4 + 2 \log 2/t^2 < 5$. This contradiction proves the finite inequality, and the two sequence conclusions follow. $\square$

The hypotheses concern one specified cancellation architecture. An application to S must still produce the positive moment and the factorial-size bound from the series, and must show that the channels through $2t^2$ vanish for the resulting family. The theorem restricts that construction and supplies no irrationality argument.

Theorem 2.4 raises the asymptotic constant in the same estimate.

Corollary 5.6 (asymptotic radius constant). *Let $M(t), R(t)$ satisfy $M(t) > 0$, $L_{2t^2} \mid M(t)$ and $M(t) < (R(t) + 1)! - 1$ for all sufficiently large t . Then*

$$\liminf_{t \rightarrow \infty} \frac{R(t) + 1}{t^3} \geq \frac{16}{9}.$$

Proof. Put $r = R(t) + 1$, so $L_{2t^2} \leq M(t) < r!$ and $\log L_{2t^2} < r \log r$. Theorem 2.4 at $N = 2t^2$ gives, for every $\varepsilon > 0$ and all large t ,

$$\log L_{2t^2} \geq \left(\frac{2\sqrt{2}}{3} - \varepsilon\right) 2\sqrt{2} t^3 (2 \log t + \log 2) \geq \left(\frac{16}{3} - \varepsilon'\right) t^3 \log t.$$

Suppose $r \leq ct^3$ for arbitrarily large t , with $c < 16/9$ fixed. On that subsequence $r \log r \leq ct^3 (3 \log t + \log c) = (3c + o(1))t^3 \log t$, and $3c < 16/3$ contradicts the lower bound for large t . $\square$

This corollary is asymptotic, and it does not give the strict finite inequality $R + 1 > \frac{16}{9}t^3$ at every large t . The finite logarithmic constraint of the channel argument is compatible with equality $9(R + 1) = 16t^3$ for $t \geq 4$, so that constraint alone cannot force a strict finite endpoint at $16/9$. The two statements are separate, and Corollary 5.6 rests on the ordinary proof of Theorem 2.4; the finite-block Lean source does not carry the asymptotic deduction. Lean checks the corollary as [asymptotic radius constant liminf](#).

The congruence (18) also explains the prime-channel corrector.

Theorem 5.7 (two-term prime channel corrector). *Let $p \geq 3$ be prime and let $c_{p-1} = p$, $c_p = -1$, with every other coefficient zero. Then $M(c) = 0$, $V_p(c) = p! - 1$, and $V_d(c) = 0$ for every $d \geq 2$ with $d \neq p$.*

Proof. Since $p \geq 3$, both indices $p - 1$ and p lie in the prescribed coefficient domain $i \geq 2$. The moment is $p(p - 1)! - p! = 0$. For $2 \leq d < p$ the quotient identity $\lfloor (p - 1)/d \rfloor = \lfloor p/d \rfloor$ holds, since $d \nmid p$, so the two channel weights carry the same power of $d!$ and the channel evaluates to $p(p - 1)!/(d!)^{\lfloor p/d \rfloor} - p!/(d!)^{\lfloor p/d \rfloor} = 0$. For $d > p$ both indices lie below d , so both weights are the plain factorials and the same cancellation occurs. At $d = p$ the weights are $(p - 1)!$ and $p!/p! = 1$, giving $p(p - 1)! - 1 = p! - 1$. $\square$

For $p = 2$, the same auxiliary identities would require $c_1 = 2$. They do not define an admissible corrector on the domain $i \geq 2$, since every vector supported on that domain has $c_1 = 0$.

At zero cost in the moment this supplies a unit in the p -channel: adding an integer multiple of the corrector to any candidate kernel shifts the p -channel by multiples of $p! - 1$ and leaves the moment and every other channel fixed. For every channel rank and every prescribed cutoff there is a factorial-grid kernel together with a remote prime-corrector pair entirely beyond that cutoff, with all requested low channels zero, nonzero moment, and residual in $[-1/2, 1/2]$. What is missing is strict nonvanishing: nothing here rules out the rounded residual being exactly zero, and no cofinal family with a strictly nonzero rounded residual has been produced.

6 Finite denominator exclusions

Two finite computations constrain a hypothetical denominator q in $S = a/q$, and they constrain different features of it:

$$q \nmid 299999!, \quad q \geq 2^{39990} > 10^{12038}. \quad (23)$$

The size bound holds for every integer a and positive natural q with $S = a/q$; no reduced-ness assumption is needed. Its evidence is the exact continued-fraction computation described below. The stronger candidate [finite-size Lean certificate](#) is not cited as kernel verification: its source records that the computation has not yet been kernel checked. The factorial-grid exclusion $q \nmid 299999!$ is supported separately by the external GMP computation; its full Lean certificate also remains an outstanding formal obligation.

Neither condition implies the other. A prime between 299999 and 599998 fails to divide 299999! but is smaller than 2^{39990} , whereas 299999! satisfies the size lower bound but fails the nondivisibility condition. High powers of small primes further show why nondivisibility is not a smoothness exclusion.

The first exclusion comes from an exact interval carry census. An independently implemented GMP integer computation certifies all 299998 carry cells for $3 \leq m \leq 300000$ with a scale of $2^{5025679}$ and 96 guard bits, using no floating-point arithmetic; at every step the outward interval is proved to lie inside one half-open unit cell before the carry is recorded. Its unit carries occur exactly at

$$52, 591, 1030, 1407, 1438, 2164, 4258, 10991, 21236,$$

so $b_{300000} \neq 1$. Theorem 3.1 then gives $q \nmid 299999!$ and in particular $q \geq 300000$. The receipt is `verification/erdos68-strict-successor.json`, schema `erdos68_strict_successor_exact_interval_v1`, payload digest `044ef27a60f7294c39b8279b7f04e22e1234a942e2862e9e8c64eb66505af29b`, recording the parameters above, the digests of the driver and backend used on its own run, the event-trace digest and the final enclosure. The source packet names the driver `scripts/check_erdos68_strict_successor.py` and backend `scripts/check_erdos68_strict_successor_gmp.cpp`. This revision records a fresh full-range carry replay. The published driver digest is `f25b5bd8ffbfd66bdbb42ca6a07a936aab9b4eb7d3a57446725be55acb642191` and the backend digest is `43308ddc3902dd537b789ae6e0648054a0d752adf07e5d6e1d829811e9342083`; the payload digest is `044ef27a60f7294c39b8279b7f04e22e1234a942e2862e9e8c64eb66505af29b`. The event-trace and unit-carry certificate match the previously retained receipt. The inherited independent replay at $m \leq 4000$ reproduces the unit-carry prefix 52, 591, 1030, 1407, 1438, 2164 with a non-unit carry at the endpoint, which gives $q \geq 4000$ on its own. The census is a separate computation from the Lean development, and the implication it feeds is the kernel-checked one.

The first exclusion can also be written $\kappa(q) \geq 300000$, where $\kappa(q) = \min\{r \geq 1 : q \mid r!\}$ is the factorial index used by Sondow [21, §3, Theorem 1]. This is not a smoothness exclusion and uses no approximation bound for e .

The second exclusion has a short integer description. Put $D = 2^{80000}$ and let N be the first integer with $N! - 1 > D$. Define

$$\ell = \sum_{n=2}^{N-1} \left\lfloor \frac{D}{n! - 1} \right\rfloor, \quad u = \ell + (N - 2) + \left\lfloor \frac{2D}{N! - 1} \right\rfloor + 1.$$

Each rounded prefix term loses less than one, and $(n + 1)! - 1 > (n + 1)(n! - 1)$ makes the tail from N smaller than $2/(N! - 1)$, so $\ell/D < S < u/D$.

There are exactly $N - 2$ rounded prefix terms. The extra $+1$ in u keeps the upper enclosure strict even if $2D/(N! - 1)$ happens to be an integer; the positive omitted tail keeps the lower enclosure strict. Here $N = 7054$ and $u - \ell = 7053$. Apply the continued-fraction algorithm to both endpoints at once, retaining a quotient only while the integer parts agree and both remainders are nonzero, and reversing the endpoint order at each inversion. There are 23449 common quotients, and the convergent denominators

$$Q_{-2} = 1, \quad Q_{-1} = 0, \quad Q_j = a_j Q_{j-1} + Q_{j-2}$$

satisfy $Q_{23448} \geq 2^{39990}$. Every rational in the open enclosure shares this initial segment, so its reduced denominator is at least Q_{23448} . Indeed, write P_j for the convergent numerators. Such a rational either equals P_{23448}/Q_{23448} or has a complete quotient $A/B > 1$ after the segment, with A and B coprime, and then equals $(P_{23448}A + P_{23447}B)/(Q_{23448}A + Q_{23447}B)$; the determinant identity $P_j Q_{j-1} - P_{j-1} Q_j = \pm 1$ makes both fractions reduced [17, §1.12(ii), (1.12.5)–(1.12.7), (1.12.20)–(1.12.21)]. The argument is about a common prefix and does not require the last convergent to lie inside the enclosure. The integer comparison $2^{39990} > 10^{12038}$ closes (23). The [integer replay](#) and its [receipt](#) record scale 80000 bits, bracket width 7053, 23449 certified partial quotients, power-of-two exponent 39990 and strict decimal exponent 12038.

Both certificates are finite. A refinement of either enlarges the excluded range and returns a statement of the same shape, and neither excludes an eventual unit-carry tail. The continued-fraction identities are classical [17, §1.12(ii)]; the enclosure, the prefix length and the exponent are outputs of the computation above.

7 The remaining arithmetic inputs

The exact target is (8), equivalently (10).

All sufficient conditions below are to be proved for the actual factorial sequence. A model obeying selected congruences, or a successful bounded computation, is not a substitute for that requirement. The routes below are sufficient conditions for it, and their quantitative hypotheses are unproved. None of them is an equivalent reformulation of the problem.

A recent framework, and a missing hypothesis. Garaev, Luca and Shparlinski's harmonic-sum estimate [20, Theorem 1 and (5)] concerns unconditioned harmonic sums, not the fibre on which $n! \equiv 1 \pmod{q}$. The value-set literature addresses different invariants. Banks, Luca, Shparlinski and Stichtenoth [25] provide historical missing-value context; Klurman and Munsch [26, Theorems 2.1–2.2] distinguish unconditional

averages from GRH-dependent improvements. Grebennikov, Sagdeev, Semchankau and Vasilevskii [27, Corollaries 1.2 and 1.4] study sizes of value and quotient sets, as do the factorial-residue and representation results of Hu [22] and Garaev and Pardo [23]. None is a noncancellation theorem for the reciprocal unit weights here. A large value set alone does not upper-bound a prescribed fibre. Hu's September preprint [24, Theorem 3.1] gives $|A| \gg \min\{M, p\}^{8/15} \mu^{-4/15}$ for distinct reciprocal-affine maps with at least M internal transitions, at most $|A|$ parameters, and nonidentity quotient multiplicity at most μ . Its incidence input is Stevens and de Zeeuw [28, Theorem 4]. A proposed transfer to a factorial-conditioned harmonic fibre must first construct such internal transitions and bound their multiplicity; neither is supplied by an unconditioned value-set estimate.

Weighted collisions and complementary residues. On the tailored moving-factor blocks, the local half of (17) is

$$\log \tilde{C}_p < \log \left(\frac{2p^2}{2p+1} \cdot \frac{\prod_{j=p}^{2p-1} j}{q} \right), \quad (24)$$

and the global half is (15). Both must hold on the same cofinal family.

The order of quantifiers is part of the proposed result: for every bound, one parameter must satisfy both inequalities. Two unrelated infinite subsequences would not prove the combined criterion. For a prime r absent from the base F_p , the contribution to $v_r(\tilde{C}_p)$ counts the levels $e \geq 1$ at which at least two gaps in I_p are divisible by r^e ; for a prime dividing F_p the base valuation must first be removed, so the layer count is $v_r(\tilde{C}_p) = \#\{e \geq 1 : h_{r,e+v_r(F_p)}(p) > 1\}$ with $h_{r,e}(p) = \#\{i \in I_p : r^e \mid i! - 1\}$. A spacing bound $h_{r,e}(p)(e+1) \leq 2p+e-2$ is available. An unweighted count of collisions does not establish (24), and reflected hits may not be discarded, since Wilson reflection produces genuine collision primes. Theorem 12 of Garaev, Luca and Shparlinski [7, arXiv v1, Thm. 12, p. 16] is a multiplicity bound and supplies no lower bound for the complementary residue in (15).

Two narrower exploratory reductions. For $F_h(X) = \prod_{j=1}^h (X + j)$, joint factorial-harmonic hits a distance h apart require $F_h(x) = 1$ and $F'_h(x) = 0$. An exact calculation gives

$$\gcd(F_{12} - 1, F'_{12}) = (X - 4626)(X - 7848) \quad \text{in } \mathbb{F}_{12487}[X].$$

Thus the earlier small-prime observation of at most one critical root is not a uniform theorem. These two roots have factorial values 442 and 6300, not 1, so the unconditioned critical-root count is genuinely a relaxation of the desired fibre count. The derivation, compressed even-gap polynomial and exact replay are in the accompanying research note.

There is also a useful distinction for the gap products in Stewart's method [8, Lemma 2, (21)–(24)]. Products from adjacent gaps $i < j < k$ are unequal: equality $j!/i! = k!/j!$ would make $k!/i!$ a square, contrary to the classical non-power theorem [31, Theorem 1]. Selecting two short gaps with nearly equal lengths does not preserve

adjacency. The wider equal-product problem has a separate history [29]; the doubled-length theorem discussed by Nguyen Xuan Tho [30, Theorem 1] is another special case, not arbitrary near-equal-length nonvanishing. No stronger lcm exponent is inferred from these local observations.

Repeated-support valuation amplification. Write $\Delta_n = u_n/v_n$ in lowest terms, let B_n be the part of $n! - 1$ supported on primes occurring in an earlier gap, and put

$$A_n = \prod_{\substack{r|B_n \\ v_r(v_n) < v_r(n!-1)}} r^{v_r(n!-1)}.$$

What is established is $A_n \mid v_{n+1}$ and, when $A_n > 1$, $u_{n+1} \not\equiv 0 \pmod{A_n}$. A sufficient quantitative goal, at cofinally many genuinely nonterminal indices n , is the following with $m = n + 1$ and $d = A_n$:

$$((m+2)m! - 2)v_m \leq m^2(m! - 1)(u_m \bmod d). \quad (25)$$

It implies the upper alternative of (11). A nonzero least representative can equal one while the modulus is enormous, so divisibility and logarithmic mass do not by themselves bound that representative. A lower bound for $\log A_n$, and an infinitude of square lifts, are separate possible inputs, and each needs a further implication to reach (25).

Lower-interval escape. Prove (11) at arbitrarily large indices; restricting to primes is sufficient. The upper alternative is exactly $((m+2)m! - 2)v_m \leq m^2(m! - 1)u_m$. Replacing u_m by its least nonnegative residue modulo a specified divisor of v_m gives a stronger sufficient test. Congruence recurrences alone do not count: synthetic models satisfy the available congruences while remaining in the unit-carry branch, and a zero canonical digit is a different event from membership in this Archimedean interval.

Doubled-prime branch failure. For an odd prime p , the carry range and $Z_{2p} = 2pZ_{2p-1} + 1 - b_{2p}$ give

$$p^2 \mid Z_{2p} \iff \begin{cases} b_{2p} = 1 \text{ and } p \mid Z_{2p-1}, \text{ or} \\ b_{2p} = 1 + p \text{ and } p \mid 2Z_{2p-1} - 1. \end{cases} \quad (26)$$

Reduction modulo p forces $b_{2p} \equiv 1 \pmod{p}$, and the two displayed values are the only possibilities in $[-1, 2p - 1]$; division by p gives the predecessor conditions. For rational $S = a/q$ and $p > q$ the tail estimate gives $Z_{2p} = (2p)!S$, which is divisible by p^2 . Ruling out both branches for infinitely many p therefore proves irrationality. The two conditions must be coupled; controlling only the predecessor residue or only the carry does not meet the hypotheses.

The two branches are disjunctive alternatives for divisibility. To force its failure one must rule out both at each chosen prime, not rule out one branch on each of two different subsequences.

A remote Cramer residual. Put $s_n = ((n+2)!)^2$ and $i_j = (t+j)s_n$ for $0 \leq j \leq n+1$. Form the integer matrix A with moment row $(i_j!)_j$ and channel rows $(W_{d,i_j})_j$ for $2 \leq d \leq n+2$, let c be the cofactor vector of the moment row, and let N_d be the determinant obtained by replacing that row with $(W_{d,i_j})_j$. Cofactor expansion gives $M(c) = \det A$, $V_d(c) = N_d$, and $V_d(c) = 0$ for $2 \leq d \leq n+2$. By (18) each $(V_d(c) - M(c))/(d! - 1)$ is an integer and vanishes for $d > \max_j i_j$, so

$$\mathcal{R}_{n,t} := \sum_{d > n+2} \frac{N_d}{d! - 1} = \det(A) S + K_{n,t}, \quad K_{n,t} \in \mathbb{Z}.$$

Seek a family with $\min_j i_j = t s_n \rightarrow \infty$ and $\mathcal{R}_{n,t} \notin \mathbb{Z}$. Such a family proves irrationality: for a fixed denominator q , every entry of the moment row is divisible by q once the support is sufficiently remote, so $q \mid \det A$. Unbounded pairs (n, t) do not supply that remoteness, because $t = 0$ anchors the least support index at zero. The eventual identity $N_d = \det A$ controls the tail and gives no control over the finite intermediate block, which changes sign; any gcd-of-minors certificate must specify a positive gcd, its rank hypotheses, and its connection to the displayed residual.

Criteria from the literature that do not apply. Duverney's Theorem 3.1 assumes the quadratic growth condition (1.3), $cu_n^2 \leq u_{n+1} \leq c'u_n^2$, whereas $u_{n+1}/u_n^2 \rightarrow 0$ for $u_n = n! - 1$; the summability condition (3.6) of his Corollary 3.2, $\sum_n |u_{n+1}/u_n^2 - 1| < \infty$, also fails, since its summands tend to one here [10, pp. 275, 285–287]. For a single denominator sequence the rapid-growth criterion goes back to Erdős [3, Theorem 1, p. 1]; Barreto, Kang, Kim, Kovač and Zhang treat products of consecutive denominators and weighted extensions [12, Thms. 2–3 and Rem. 4]. For $a_n = n! - 1$, $\log(n! - 1) = O(n \log n)$ makes $(n! - 1)^{1/\psi^n} \rightarrow 1$ for every fixed $\psi > 1$, so neither Erdős's hypothesis $\limsup_n a_n^{1/2^n} = \infty$ nor the growth hypotheses of their Theorems 2 and 3 hold. These criteria do not decide S ; the useful content is the target these papers identify: a low-height clearing subsequence, or enough exact cancellation in the least common multiple. Theorem 2.4 bounds that cancellation from one side and Theorem 1.2 describes it from the other.

No irrationality conclusion is obtained here. The finite conclusion is that every rational representation $S = a/q$, $q > 0$, must satisfy both $q \nmid 299999!$ and $q \geq 2^{39990} > 10^{12038}$. The remaining task is a cofinal estimate, not a larger finite census.

Sources and evidence

The formal counterparts below are checked by the Lean 4 kernel against Mathlib at source snapshot 92b88dc1bbe0, with no sorry, added axioms, or unchecked evaluation. Theorem 2.4 and Corollary 5.6 are ordinary proofs given in full above, and Lean checks them as [common denominator growth liminf](#) and [asymptotic radius constant liminf](#). The finite-block inequality is the separate Lean source [finite terminal-block inequality](#). The two prefix cancellations, the index-52 example and the continued-fraction enclosure are finite integer calculations with the procedures displayed above. The carry census through 300000 is a separate exact-interval computation with the receipt named in §6.

Statement	Formal counterpart
Theorem 1.2	maximal-power survival, with the residue formula at line 129 and the denominator form at line 41
Proposition 1.3	cofinal private hits; reflection at line 3139
Theorem 3.1	strict-successor misses; denominator exclusions at line 812 and line 836
Proposition 3.2	companion orbit
Proposition 3.3	lower-interval normal form; finite window at line 122, cofinal consumer at line 145, classification (12) at line 39, radius at line 158
Theorem 3.4	family boundary; escape form at line 156, member $t = -1$ at line 175, and the irrationality of e at line 211
Theorem 4.1	global complementary residue; coprimality (14) at line 4209
Equation (17)	unit-factor scale split, floor identity at line 4701, consumer at line 1574
Fixed-owner absorption	absorbed owner
Theorem 5.1 and (18)	integral channel weight, exact cancellation
Theorem 5.5	finite radius bound; sequence form at line 1103, little- o form at line 905, method ceiling ordinary in this record (sharp_radius_satisfies_square_log_constraint is not in this checkout)
Prime channel corrector	remote reduction, residual identity at line 1559
Equation (26)	doubled-prime criterion
Canonical factorial digits	termination equivalence
Amplification modulus	divisibility, nonvanishing at line 5572

Attribution. Wilson's theorem and the Wilson reflection identity are classical, the latter recorded by Stewart [8, p. 462, (4)]. The factorial-digit termination criterion goes back to Cantor [5]; Koepf and Schmersau prove its irrationality direction for digits that are not eventually maximal [9, Example 3.2, p. 121], and Galambos treats rationality criteria for Cantor series [6, Ch. II, §2.1]. The additions here are the telescope $C = S - e + 2$, the identification of the digit value $m - 2$, and the transport to the shifted family. The tail-integrality step is the argument of Hančl and Tijdeman [11, Lemma 2.1 and the following remark, p. 385]. The multiplicity bound is Garaev, Luca and Shparlinski's [7], and the lcm deduction from it is not theirs. The divisibility in Lemma 2.2 is the case $P = -1$ of the relation used in the proof of Lemma 5 of Luca and Shparlinski [13] and at display (2.5) of Lai [14]. The nonvanishing cutoff for polynomial shifts is Lemma 3 of [13], restated with the bound $n! + P(n) > 1$ in [14, Lemma 2.1]. The survival criterion is

a specialisation of Louwsma and Martino’s valuation formula [4, Lemma 4.1, p. 10]. The single-denominator growth criterion is Erdős’s [3, Theorem 1, p. 1], and the continued-fraction identities are those of [17, §1.12(ii)]. No antecedent was located for the Wilson cofinality packaging, the finite radius bound or Theorem 2.4.

A Guide to the formal sources

The public `ErdosProblems.Erdos68` package carries the checked source for this note. The declarations below are the dependency chain behind the statements above, together with the subsidiary results developed alongside them; they are pinned to the formal-source commit named at the start of this note.

- `fac floor rat eq cleared`
- `canonical digit eq zero of rational`
- `canonical digit eq floor mul remainder`
- `canonical digit lt radix`
- `canonical remainder recurrence`
- `factorial expansion partial`
- `zero remainder forces zero digit tail`
- `canonical remainder eq zero of zero digit tail`
- `exists rat eq of canonical remainder eq zero`
- `factorial grid plateau`
- `irrational factorial gap series of cofinal nonunit carries`
- `sixty seven le rational denominator`
- `factorial gap zero branch iff lower endpoint cylinder`
- `exists late prefix private factorial gap hit of prime product lt`
- `prime dvd factorial block normalized collision core of odd reflection`
- `one lt factorial block prime hit count of odd reflection`
- `factorial gap large prefix private power modulus dvd tailored block private quotient`
- `factorial gap large prefix private primes unit factor pair tailored block`
- `consecutive unit carries iff positive offset le den`
- `two step den mul transition normalizers`
- `adjacent unit carry window den eq two step den`
- `adjacent unit carry window offset eq two step reduced numerator`
- `factorial moment eq factorial pow mul channel numerator band`

- factorial moment eq zero of channel numerator eq zero band
- factorial moment eq factorial mul channel numerator first band
- exists index ge two mul of factorial moment ne zero of channel eq zero
- square subsequence radius cubic lower
- no eventual square subsequence cubic upper
- channel event eq zero of not dvd
- lambda34 residual bounds
- tendsto factorial gap succ div sq
- not summable abs factorial gap succ div sq sub one
- tendsto factorial gap rpow one div two pow
- pairwise collision core insert gcd lcm
- collision core insert gcd lcm
- collision core div base eq pairwise collision core div gcd
- collision core div base factorization
- collision core div base mul denominator lcm dvd denominator prod
- factorial block fixed pair private quotients eq one
- factorial block normalized collision core le gap prod div gap lcm
- factorial block normalized collision core factorization
- factorial block prime power le gap pow of two hits
- factorial block prime hit pair distance gt exponent
- factorial block prime power hit count mul succ le
- factorial block normalized collision factorization add base lt prime
- factorial block normalized collision factorization add base lt block diameter
- factorial block prime sq gt pred of dvd normalized collision core
- factorial block factor projection lcm eq private modulus
- factorial block complementary factor pair floor le global
- factorial block normalized collision core factorization eq repeated hit layer count
- factorial block normalized collision core factorization eq truncated repeated hit layer count
- factorial block prime power dvd normalized collision core iff one lt hit count of end-point lt base

B Extended record

The short note above carries the advertised results with complete proofs. This part keeps the surrounding mathematics: the digit kernel the carry and orbit criteria run on, the exact interfaces of the collision-core reduction, the finite certificates, the superseded deductions, and the routes that were tried and closed.

B.1 The canonical factorial digit kernel

Write $\theta_0 = \{x\}$ and, for $m \geq 1$,

$$a_m = \lfloor m \theta_{m-1} \rfloor, \quad \theta_m = m \theta_{m-1} - a_m,$$

so that $\theta_m = \{m!x\}$ for $m \geq 1$. The kernel checks the floor formula, the digit bounds $0 \leq a_m < m$, the recurrence $\theta_{m+1} = (m+1)\theta_m - a_{m+1}$, the finite telescoping expansion

$$x = \lfloor x \rfloor + \sum_{m=2}^N \frac{a_m}{m!} + \frac{\theta_N}{N!},$$

and the rule that a zero remainder at one index forces every later digit to vanish. The rational direction is also checked: if $q > 0$ and $q \leq n$, then $\text{facFloor}(a/q, n) = ((n!/q) : \mathbb{Z}) a$ and the canonical digit at radix $n+1$ vanishes, so every rational input has an eventually zero expansion. The converse holds for every real input, and gives Cantor's termination criterion [5]

$$x \in \mathbb{Q} \iff a_m(x) = 0 \text{ for all large } m.$$

If all digits after index $N \geq 1$ vanish, the recurrence gives $\theta_{N+k} \geq (k+1)\theta_N$; since every remainder is below one, $\theta_N = 0$ and $x = \lfloor N!x \rfloor / N!$.

A zero canonical digit is a different event from a zero-branch hit. The reported interval data contain canonical zero digits at $m = 5$ and $m = 23$, while the zero-branch list is empty through $m = 100000$.

A second exact reformulation runs through a defect automaton. For a rational centre recurrence $F_m = mF_{m-1} + 1 + \varepsilon_m - C_m$, the kernel checks that the integer ceiling defect code equals $\lfloor m\delta_{m-1} - \varepsilon_m \rfloor$ and that $\delta_m = m\delta_{m-1} - \varepsilon_m - q_m$, with the specialisation $\varepsilon_m = 1/(m! - 1)$ written out. What is checked is the algebra of the automaton; proving that the finite-sum residual centre satisfies the premise is a separate step and is not done.

B.2 Criteria from the literature, in full

Koepp and Schmiersau prove that eventual equality between the floors of n times a partial sum and n times its limit forces irrationality [9, Thm. 1.1, p. 117]; their rational-term version obtains that equality from the integrality condition $np_n s_n \in \mathbb{N}_0$ and the strict tail bound $a - s_n < 1/(np_n)$ [9, (2.1) and (2.3), p. 118; Thms. 2.2–2.3, pp. 119–120]. For the choice they record after (2.1), the least common multiple of the reduced summand denominators, here $p_n = \text{lcm}\{k! - 1 : 2 \leq k \leq n\}$, the last two denominators already obstruct the tail bound:

$$p_n \geq \frac{(n! - 1)((n-1)! - 1)}{n-1},$$

because $\gcd(n!-1, (n-1)!-1) = \gcd((n-1)!-1, n-1) \leq n-1$. For $n \geq 4$ the first omitted summand $1/((n+1)!-1)$ then already exceeds $1/(np_n)$, so this p_n cannot satisfy their tail hypothesis. Cancellation in the reduced prefix denominator could give a smaller scale, and proving enough cancellation is another form of the present denominator problem.

Duverney's Theorem 3.1 assumes the quadratic growth condition (1.3), $cu_n^2 \leq u_{n+1} \leq c'u_n^2$, while $u_{n+1}/u_n^2 \rightarrow 0$ for $u_n = n! - 1$ [10, (1.3), p. 275; Thm. 3.1, pp. 285–286]; his Corollary 3.2, which allows signs $a_n \in \{-1, 1\}$, requires the summability condition (3.6), $\sum_n |u_{n+1}/u_n^2 - 1| < \infty$, whereas those summands tend to one [10, Cor. 3.2, p. 287]. Both the ratio limit and the nonsummability are kernel-checked. The ratio proof divides numerator and denominator by $(n!)^2$; the resulting terms tend to zero while $(1 - 1/n!)^2$ tends to one.

For a single denominator sequence, Erdős proved that $\sum_k 1/n_k$ is irrational when $\limsup_k n_k^{1/2^k} = \infty$ and $n_k > k^{1+\varepsilon}$ eventually [3, Theorem 1, p. 1]. Barreto, Kang, Kim, Kovač and Zhang treat products of consecutive denominators and weighted extensions [12, Thms. 2–3 and Rem. 4, pp. 2–5]. For $a_n = n! - 1$ the root-growth expressions tend to one, $(n! - 1)^{1/2^n} \rightarrow 1$, so neither the limsup hypothesis of Erdős nor the growth hypotheses of their Theorems 2 and 3 hold; the limit is kernel-checked through the bound $0 \leq \log(n! - 1) \leq n^2$. Their proof of Theorem 3 uses the classical criterion, which they trace to Fourier's proof that e is irrational, that a rational sum of nonnegative rationals with infinitely many positive terms admits no prefix-clearing integers D_N with $\liminf_N D_N r_N = 0$, where r_N is the tail after N terms [12, Lem. 8, p. 6]; their Proposition 12 produces such integers under the hypotheses of that theorem [12, pp. 9–12]. The ordinary product of the factorial-gap denominators is far too large for that estimate, so a transfer needs a low-height clearing subsequence, or enough exact cancellation in their least common multiple.

Dividing the strict-successor recurrence $Z_m = mZ_{m-1} + 1 - b_m$ by $m!$ and telescoping gives the exact finite identity

$$\frac{Z_M}{M!} = \frac{Z_2}{2!} + \sum_{m=3}^M \frac{1 - b_m}{m!},$$

so the carry defects $1 - b_m$ are genuine factorial-series coefficients. Hančl and Tijdeman classify the rational sums with polynomial coefficients [11, Thm. 3.1 and Cor. 3.1, pp. 390–391]; their denominator is the cumulative linear product $\prod_{n \leq N} (an + b)$, and the individual number $N! - 1$ does not occur. The criterion of Oppenheim that they reproduce [11, Lem. 2.2, p. 385] states that, for coefficients c_n with $|c_n| < an + b$ eventually and $\liminf_n |c_n|/n = 0$, the sum is rational exactly when $c_n = 0$ eventually. For the carry defects $c_m = 1 - b_m$ the first condition holds, since $-1 \leq b_m \leq m - 1$, and the second is not established; the needed cofinal equivalence is Theorem 3.1, which is proved directly.

B.3 A superseded deduction

Before the elementary segment argument of the short note, the record derived a weaker exponent from an external multiplicity theorem. Put $Q_N = \prod_{2 \leq n \leq N} (n! - 1)$. For an odd prime r , let m_r count the indices $2 \leq n \leq N$ with $r \mid n! - 1$; such an index satisfies

$n < r$. The factorial-congruence multiplicity estimate of Garaev, Luca and Shparlinski [7, arXiv v1, Thm. 12, p. 16], applied to the residue $a = 1$ on $1 \leq n \leq \min(N, r-1)$, gives $m_r \ll N^{2/3}$; the prime 2 divides none of these factors. Writing $E_r = \max_{2 \leq n \leq N} v_r(n! - 1)$,

$$\log Q_N = \sum_r \sum_{n=2}^N v_r(n! - 1) \log r \leq (\max_r m_r) \sum_r E_r \log r \ll N^{2/3} \log L_N,$$

and Stirling summation gives $\log Q_N \asymp N^2 \log N$, whence $\log L_N \gg N^{4/3} \log N$. Theorem 2.4 supersedes this: its exponent is $3/2$, its constant is explicit, and it uses no external theorem. The deduction is retained because it is the only place a multiplicity bound enters the record.

The primitive lcm divisibility survives cofactor removal: factorial valuations do not remove the channel obstruction once every common cofactor divisor has been removed. A corank-one cofactor and determinant construction for such primitive kernels has not been formalised, and the divisibility theorem does not establish it.

B.4 Plateaux, first exit, and the finite denominator ladder

A second argument works on the rational grid in place of the channels. Let H be a partial sum and q a candidate denominator. Writing $qH = k + r$ and $q(S - H) = u$, the next q^{-1} grid point $(k + 1)/q$ lies below S exactly when $1 \leq r + u$. The factorial plateau theorem states that if $H < G \leq S$, if $n!G$ is integral, and if $n!(S - H) < 1$, then the strict successor of $n!H$ and the canonical floor of $n!S$ are the same grid integer. Two rigidity statements follow: consecutive plateau floors, scaled by the next radix, force the canonical digit to vanish; and any first-exit offset $\delta \in [0, 2)$ with carry $b = -\lfloor \delta \rfloor$ has $b \in \{0, -1\}$, so the exit has exactly two alternatives.

The first-crossing argument continues to a denominator bound. For a rational grid level G with first crossing τ by the literal partial sums, write $G - H_{\tau-1} = a/v$ with $a, v > 0$. Then $v \geq \tau! - 1$, and on the -1 exit branch $v \geq \tau!(\tau! - 1)$. No coprimality hypothesis on a and v is needed.

There is also a direct obstruction at prime indices. With Δ_m as in the short note, the kernel checks for $m \geq 3$ the criterion

$$m \mid Z_m \iff 1 + \frac{1}{m! - 1} < m\Delta_m \leq 2 + \frac{1}{m! - 1},$$

and if $S = a/q$ with $q > 0$, then $p \mid Z_p$ for every prime $p > q$, so one exact missed prime p gives $q \geq p$. Exact kernel reduction gives $11 \nmid Z_{11}$, hence $q \geq 11$; exact rational normalisation gives $60 \nmid Z_{60}$, $64 \nmid Z_{64}$ and $67 \nmid Z_{67}$, and the prime index 67 gives the checked bound

$$S = \frac{a}{q}, q > 0 \implies q \geq 67.$$

The exact-interval census of the short note replaces 67 by 300000.

There is a finite peeling identity. For $x \neq 0, 1$ and $K \geq 0$,

$$\frac{1}{x-1} = \sum_{j=1}^K \frac{1}{x^j} + \frac{1}{x^K(x-1)}.$$

When $x = k!$ and a chosen factorial scale is divisible by $(k!)^K$, the scaled finite sum is integral and only the last term retains the factor $k! - 1$ in its denominator. The identity isolates one residual fraction before exact bounding, and it supplies no cofinal family of nonzero residuals.

One proposed strengthening is false and is recorded as false: the divisibility $(m! - 1) \mid \text{den}(V_m)$ fails at the reported strict events $m = 52$ and $m = 591$. Only the Archimedean first-crossing lower bound survives.

There is a second, more arithmetic mechanism at doubled prime indices, stated as (26) in the short note. The formal theorem is not restricted to hand-reduced indices; it is the general unique-slot prime-power criterion specialised to the literal prefixes.

B.5 The collision-core interfaces

Beyond the definitions used in the short note, the record keeps the exact interfaces of the collision-core reduction.

Write $D(S) = \text{lcm}_{i < j, i, j \in S} \text{gcd}(d_i, d_j)$, with $D(S) = 1$ for an empty or singleton family. For each prime r , its valuation is the second-largest valuation among the d_i , counting missing values as zero; the denominator lcm has the largest valuation. This proves $D(S) \mid \text{lcm}_{i \in S} d_i \mid \prod_{i \in S} d_i$ prime by prime.

The collision core has an exact incremental law. For the positive factorial-gap denominators, adjoining a new index $a \notin S$ gives

$$D(S \cup \{a\}) = \text{lcm}(D(S), \text{gcd}(d_a, \text{lcm}_{j \in S} d_j)),$$

since finite-family gcd and lcm distributivity collapses the lcm of all pairwise gcds against d_a to a single gcd. The same formula holds after adjoining the distinguished base, so each step needs only the old denominator lcm and the old core, with no pairwise rescan.

There is an exact product and lcm bound. If $\tilde{C}(S)$ is the core after cancelling a positive distinguished base, $L(S) = \text{lcm}_{j \in S} d_j$ and $P(S) = \prod_{j \in S} d_j$, then $\tilde{C}(S)L(S) \mid P(S)$, hence $\tilde{C}(S) \leq P(S)/L(S)$. For the factorial block this specialises to

$$\tilde{C}_p \leq \frac{\prod_{n \in I_p} (n! - 1)}{\text{lcm}_{n \in I_p} (n! - 1)},$$

an exact bridge from lower estimates for the factorial-gap lcm to upper estimates for the normalised core. It does not close the local scale bound.

The base cancellation is exact prime by prime. With $F_p = (p - 1)!$, $D_p = D(I_p)$ and $C_p = \text{lcm}(F_p, D_p)$, as in the main record,

$$\tilde{C}_p = \frac{C_p}{F_p} = \frac{D_p}{\text{gcd}(F_p, D_p)}, \quad v_r(\tilde{C}_p) = \max\{0, v_r(D_p) - v_r(F_p)\}.$$

For $e > 0$, $r^e \mid \tilde{C}_p$ exactly when the pairwise core carries $r^{e+v_r(F_p)}$, which in the block forces two distinct gaps to be divisible by that higher power, by the second-largest-valuation formula. If $i < j$ are two such hits and $f = v_r(F_p)$, then $r^{e+f} \mid j!/i! - 1$. Since

$r \mid j! - 1$ forces $j < r$, $0 < j!/i! - 1 < r^{j-i}$, hence $e + f < j - i$. The sharp surviving valuation cap $v_r(\tilde{C}_p) + v_r(F_p) < r$ then gives $p - 1 < r(r - 1) < r^2$ for every support prime, and $\tilde{C}_p$ is coprime to $k!$ whenever $k(k - 1) \leq p - 1$. This removes every factorial channel below the moving square-root cutoff, and it does not bound the aggregate product of the remaining large prime powers.

For collision estimates that already provide an upper-half hit, no exponent is lost to normalisation: if r divides a displayed gap at some $n \geq p$, then $r \nmid F_p$ and $r^e \mid \tilde{C}_p \iff r^e \mid C_p$ for every $e > 0$. This has an exact incidence-count form,

$$r^e \mid \tilde{C}_p \iff 1 < \#\{i \in I_p : r^e \mid i! - 1\},$$

so a source estimate giving at most one r^e -hit deletes that exponent and yields $v_r(\tilde{C}_p) < e$. The local aggregation is exact:

$$v_r(\tilde{C}_p) = \#\{e \in [1, r - 1] : 1 < \#\{i \in I_p : r^e \mid i! - 1\}\},$$

and for an endpoint prime with $2p - 1 < r$ the range truncates to $e \in [1, 2p - 4]$. The same equivalence holds under the exact condition $r \nmid F_p$ in place of the endpoint inequality, which covers the entire moving prime range at and above the block parameter; at $e = 2$ it gives a conditional squarefree conclusion $v_r(\tilde{C}_p) \leq 1$.

The local load has a distance-sensitive witness. If r is prime, $e > 0$ and $r^e \mid \tilde{C}_p$, there are $i < j$ in I_p with $r^{e+v_r(F_p)} \mid i! - 1$, $r^{e+v_r(F_p)} \mid j! - 1$ and $r^{e+v_r(F_p)} \leq j^{j-i}$; consequently $(2p - 1)^d < r^{e+v_r(F_p)}$ forces $d < j - i$. The spacing hypothesis is discharged internally: any two r^e -hits $i < j$ satisfy $e < j - i$, because $r \mid i! - 1$ already forces $j < r$ and the gap-power inequality converts that size relation into strict separation. The pairwise form is stronger than the selected-witness form: for arbitrary r, e and any displayed hits $i < j$,

$$r^e \mid i! - 1, \quad r^e \mid j! - 1 \implies r^e \leq j^{j-i},$$

so every prime-power hit layer is an e -separated subset of the block and

$$(e + 1) \#\{i \in I_p : r^e \mid i! - 1\} \leq 2p + e - 2.$$

The unweighted packing step is therefore complete. Globally,

$$r \mid \tilde{C}_p \implies v_r(\tilde{C}_p) + v_r(F_p) < 2p - 3$$

for every prime r and $p \geq 2$, so even primes already present in the normalisation base pay for their base valuation inside the same block-diameter budget. What remains is to bound the layer counts uniformly as r and p move, multiply the surviving contributions, and still close the complementary-residue coordinate.

The squarefreeness premise is not proved. An exhaustive modular scan through $r \leq 2,000,000$ and $n \leq 240$ found four individual square hits and no prime with two such hits. All 498,501 pairs $2 \leq a < b \leq 1000$ have squarefree $\gcd(a! - 1, b! - 1)$, and the aggregate squarefree-collision scan through $p = 499$ stays below 0.374 of the upper-descending-factorial logarithmic scale. These are finite exact computations. They carry neither theorem authority nor an asymptotic incidence bound.

A finite variant of the cofinal private supply compares the product of a chosen set of primes, each at least 5, with $\prod_{2 \leq k \leq B} (k! - 1)$; if the prime product is larger, at least one chosen prime has no hit through B , while Wilson still bounds its least hit by $q - 2$. Wilson reflection limits what a linear-size divisor can be assumed to be: if n is odd, $n < q$, and $q \mid n! - 1$, then $q \mid (q - n - 1)! - 1$, and when both indices lie in one block with the reflected hit earlier, equivalently $q < 2n + 1$, the repeated hit survives predecessor-factorial normalisation and its full-block incidence count exceeds one.

Stewart states that for every $\varepsilon > 0$ there are infinitely many odd n whose least prime factor q of $n! - 1$ satisfies

$$n < q < \left(\frac{\sqrt{145} - 1}{8} + \varepsilon \right) n,$$

estimate (9) of Theorem 1 being stated for $n! + 1$ [8, p. 463] and transferred to $n! - 1$ in the text [8, p. 464]. The source controls q relative to the later index n and supplies no control relative to the private first-hit index, so it is collision-core input. The missing private-anchor and global product estimates are elsewhere.

One selected prime q already furnishes the coprime factor pair $(1, q)$: its projection moduli are R_p and R_p/q , whose least common multiple is R_p , so no second selected prime is needed and there is no hidden equality-or-disagreement branch in that specialisation. The underlying projection rigidity is general: if the endpoint numerator Z is congruent to a weighted numerator T modulo R , then every divisor Q of R with $Z \leq B < Q$ satisfies $T \bmod Q = Z$, so two divisors above B with unequal projected residues exclude that bounded endpoint, and unequal projections force $\min(Q_1, Q_2) \leq T$.

B.6 Finite channel and carry certificates

The finite-support vector $\lambda = 2e_3 - e_4$ has, by kernel check, $V_2(\lambda) = 0$, factorial moment -12 , $V_3(\lambda) = -2$, $V_4(\lambda) = 11$, and $V_d(\lambda) = -12$ for every $d \geq 5$. Under the exact rational tail enclosure $1/119 < \Theta_4 < 1/50$ its residual lies strictly between $-93/575$ and $-309/13685$, so it is nonzero and subunit.

In the enlarged coefficient space of §5, exact integer computation verifies the vectors K_D for every $2 \leq D \leq 12$: channels 2 through D vanish, the factorial moment is L_D , and the coefficient content is one. They lie outside the space of vectors supported on $n \geq 2$, since L_D is odd and every such vector has even moment. At $D = 9$,

$$L_9 = 31540008254514077395, \quad a_9 = [e_1]K_9 = -3902884074990939115.$$

Since $U_{11} = T_{11}$ has $u_{11} = 0$, the vector $K_9 - 9553024718754 U_{11}$ keeps the coordinate a_9 and has content one. By Theorem 5.4 its residual is $L_9(S - H_9) - 9553024718754$, and exact rational arithmetic with the tail bound $\sum_{n \geq 36} 1/(n! - 1) < 2/(36! - 1)$ places it strictly between $1353/100000$ and $1354/100000$. A different example is supported on $n \geq 2$: the vector $c = (-40, 55, -10, 1)$ on the support $(3, 4, 5, 6)$ annihilates channels 2 and 3, has moment 600, and satisfies

$$0.09925341997208298 < \mathcal{R}(c) < 0.09925341997208300,$$

which excludes denominators dividing 600.

There are two different computations at the same endpoint. A separate interval computation reports the stronger geometric statement that no zero-branch event occurs at any $m \leq 100000$; its executable and source digest are not available, so that classification remains external finite evidence. The strict-successor census through 300000 used in the short note is a fresh exact GMP replay in this revision; §6 records the published driver, backend and payload digests and describes the separate replay through 4000. The full range is external computational evidence outside the Lean development.

B.7 Limits of fixed-coordinate arguments

- Residue vectors, their recurrences, and window widths admit synthetic all-hit blocks, so they cannot prove irrationality on their own.
- Known pointwise prime congruences, prime-dilation congruences, parity, and the exact prime coefficient formula admit a synthetic rational countermodel.
- Wilson quotients, harmonic sums, p -adic gamma identities, and factorial residues do not control the required Archimedean floor without an additional coupling theorem. Every prime-window test factors into a sharp Archimedean strict-ceiling condition and a modular divisibility condition, and the missing ingredient is the coupling between them.
- For the genus-zero product $E(z) = \prod_{n \geq 2} (1 - z/n!)$, local uniform convergence and logarithmic differentiation give $-E'(1)/E(1) = S$. The natural termwise clearing factor $Q_N = \prod_{2 \leq n \leq N} (n! - 1)$ satisfies $Q_N(S - H_N) \rightarrow \infty$, so this clearing choice yields no small positive integer linear form. The calculation concerns this clearing factor alone; Hermite–Padé systems with other denominators are left open.
- Zero-moment variations cannot create an additional fractional cancellation coordinate, and factorial valuations cannot absorb the channel lcm obstruction.
- A fixed pair of low-index private owners cannot make the projection argument cofinal, by the absorption statement of the short note.

B.8 The generic shape behind the lower-interval form

The cofinal statement in the short note is a case of the following immediate consequence of Cantor’s termination criterion. For a real x and any positive sequence c_m with $c_m < 1/m$ eventually,

$$x \notin \mathbb{Q} \iff \{(m-1)!x\} \geq c_m \text{ for arbitrarily large } m.$$

For rational x the fractional parts vanish eventually; if all large fractional parts lie below c_m , multiplying by m stays below one, so the next canonical digit is zero, and eventual zero digits force rationality. The specific target $c_m = E_m/m$ is useful because it translates into finite predecessor-gap conditions. The smallness of the target alone establishes no metric, measure or equidistribution theorem for this orbit, and none is claimed.

C How the result families fit together

Carries. Tail integrality gives the necessary eventual unit carry under rationality. Telescoping gives the converse. These are exact equivalences; the finite census supplies only bounded instances of escape.

Channels. Integral triangular elimination solves the selected channel equations. Removing the auxiliary coordinate yields a moment ideal. The finite horizon computes that ideal at fixed depth, while the residual identity explains why zero-moment corrections do not create a new fractional degree of freedom.

Common and reduced denominators. Pairwise subtraction and a terminal block give the lcm lower bound. Reciprocal-unit sums determine survival of a maximal prime-power layer after reduction. These are complementary statements about different integers.

Conditional routes. A remote residual must lie strictly between consecutive integers, or a collision-core estimate must be paired with a sufficiently large complementary gap. These are additional quantitative inputs, not consequences of divisibility alone.

Finite evidence. Carry and continued-fraction certificates exclude two incomparable sets of rational denominators. Their numerical parameters, source lineage and replay status are separately recorded. A larger certificate would not remove the need for a cofinal argument.

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